

Roll Number:.....

Thapar Institute of Engineering and Technology, Patiala
Department of Electronics and Communication Engineering

M.Tech.: MST- March 2025

Time: 2 Hours; MM: 25

PVL207: Low Power System Design

Name of Faculty: Dr. Rajneesh Sharma

Note: 1. Attempt all questions.

2. Take suitable assumption(s) if necessary.

Q.N.	Questions	Marks	CO	BL		
Q.1	A company is training a deep learning model using a high-performance computing cluster. Two GPU-based server configurations are available: <table><tr><td>1. Configuration X: Power Consumption: 2.2 kW per server Training Time per Model: 10 hours Accuracy Achieved: 97.5%</td><td>2. Configuration Y: Power Consumption: 1.5 kW per server Training Time per Model: 18 hours Accuracy Achieved: 98.2%</td></tr></table> The company pays for electricity based on total energy consumed, which configuration is more energy-efficient while maintaining model accuracy and why?	1. Configuration X: Power Consumption: 2.2 kW per server Training Time per Model: 10 hours Accuracy Achieved: 97.5%	2. Configuration Y: Power Consumption: 1.5 kW per server Training Time per Model: 18 hours Accuracy Achieved: 98.2%	[5]	CO1	L5
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Q.2	What thermal reliability challenges necessitate the adoption of low-power techniques in VLSI design? Explain these challenges in detail. Additionally, provide a brief overview of emerging low-power techniques implemented at different abstraction levels.	[5]	CO1	L2		
Q.3	Determine the switching probabilities for the outputs of a 1-bit full adder circuit, assuming that all input variables are independent and have equal probability of occurrence.	[5]	CO2	L3		
Q.4	A high-speed microprocessor and an embedded controller for IoT devices both require leakage power reduction. Would MTCMOS be an appropriate solution for both? If not, what alternative techniques should be used in each case, and why?	[5]	CO1,2	L5		
Q.5	Discuss the different sources of power consumption in VLSI design, and identify which one is most critical in today's technology. Provide a rationale for your choice	[5]	CO2	L4		